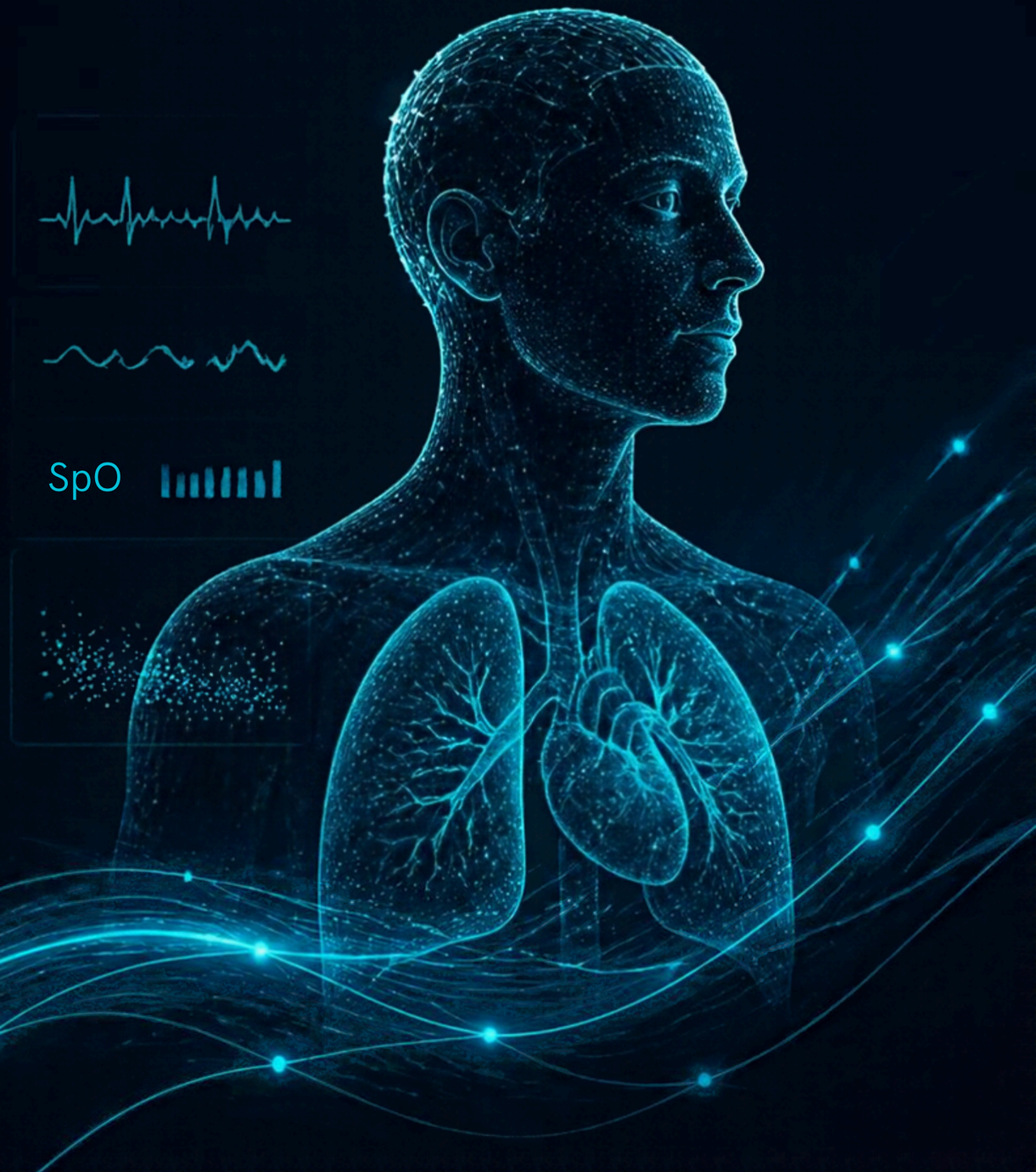
VitalTwins

Mechanistic PK/PD & Multimodal Predictive Clinical Twins

Closing the trust gap between unseen signals and clinical decisions.

TEAM CTRL win
Parth Kulkarni

Happiest Health Digital Twin Challenge 2026



The Clinical Problem

EHR records and wearable signals live in **separate worlds**.

There may be an **early warning window** before symptoms.

Clinicians need **physiological explainability**, not black boxes.

Today, **trust** is the missing link.

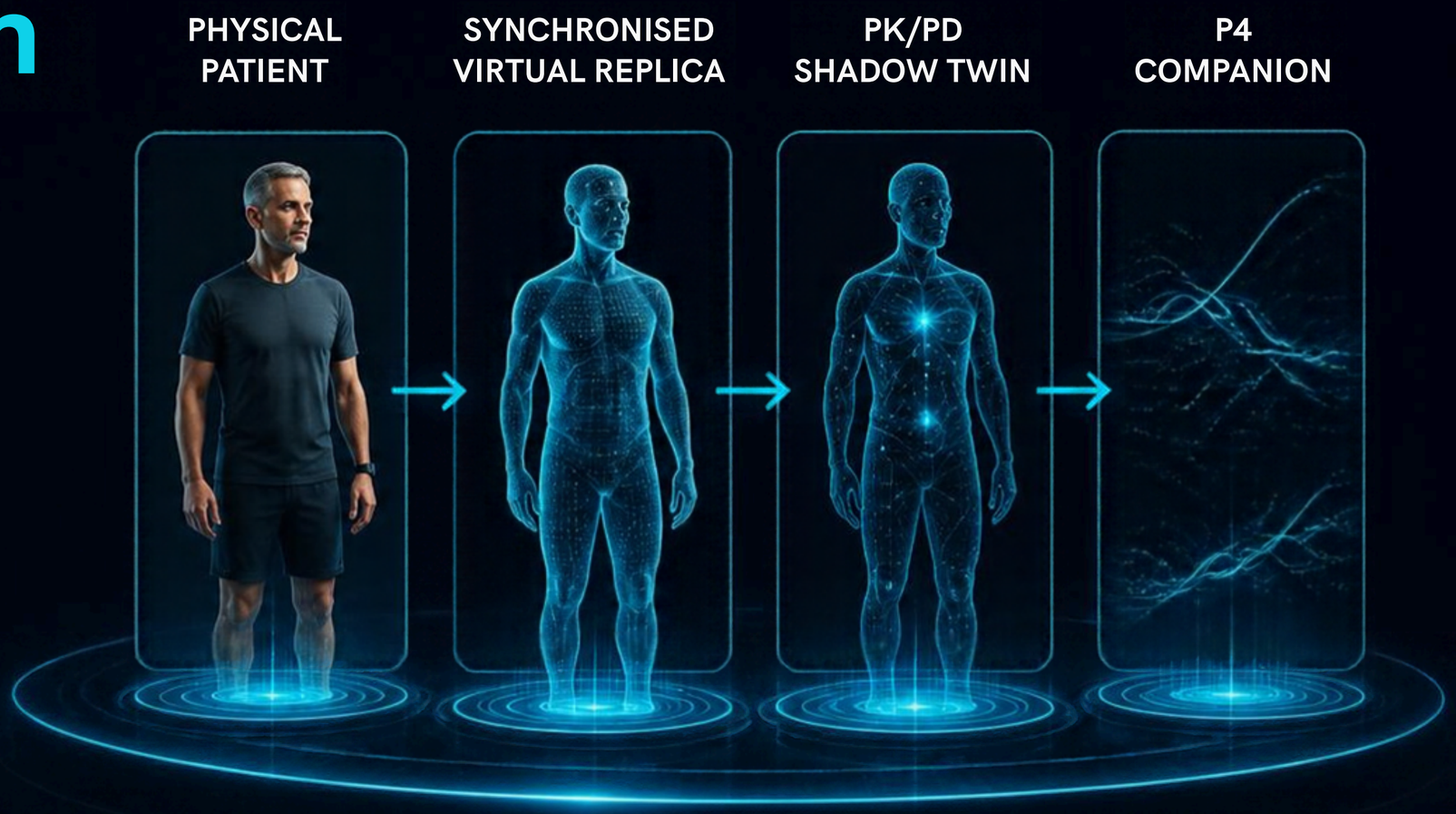
From Risk Scores to a Digital Twin

A living model of the patient,
updated in sync with real life.

Reflects both baseline
and day-by-day changes.

Explore what-ifs in a safe sandbox—
insight only, not prescriptions.

Patient-centred guidance,
built for calm, informed decisions.



COUNTERFACTUAL SIMULATION IN A SANDBOX—NOT PRESCRIBING
NOT CLINICALLY VALIDATED

System Architecture & Fusion Pipeline

SYNTHETIC EHR
structured clinical data

XGBOOST MODEL

48-HOUR WEARABLE STREAM
continuous biosignals

TEMPORAL STRESS EVALUATOR

FUSION ENGINE

multimodal integration & risk inference

XAI	explanable insights
RISK STATUS	patient-level risk assessment
FHIR R4	interoperable clinical data
SIMULATOR	what-if scenario exploration
RED ESCALATION	real-time risk escalation

Synthetic Sandbox & Fusion Math

500

synthetic profiles

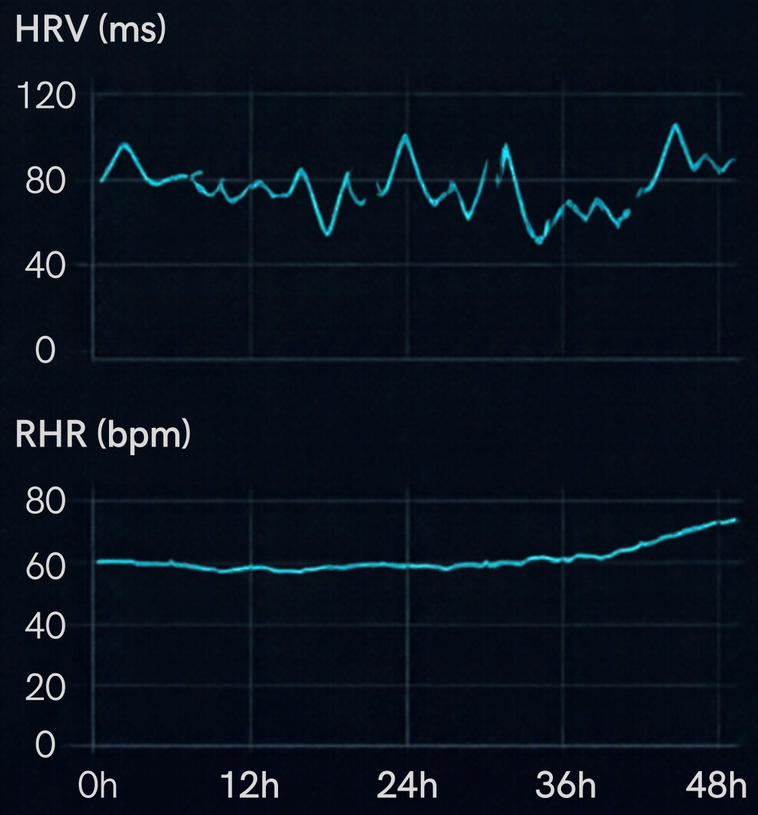
48-hour

hourly wearable stream

multimodal features

heart rate variability
resting heart rate
activity
sleep
oxygen saturation

PAT_001 EXAMPLE



FUSION EQUATION

$$RISK = w_1 \cdot HRV + w_2 \cdot RHR + w_3 \cdot ACT + w_4 \cdot SLEEP + w_5 \cdot SpO_2$$

RISK STATES

GREEN	stable low concern
YELLOW	watch moderate concern
RED	act high concern

SYNTHETIC SANDBOX RESULTS AND INVESTIGATIONAL SIMULATIONS

NOT CLINICALLY VALIDATED OR PRESCRIBING GUIDANCE.

PK/PD Shadow Twin

mechanistic models. regimen scenarios. safety guardrails.

BATEMAN PK MODEL

$$C(t) = \frac{F \cdot D \cdot k_a}{V_d\,(k_a - k)}\,(e^{-kt} - e^{-k*})$$

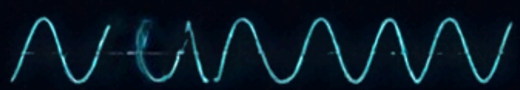
- plasma concentration
- bioavailability
- dose
- absorption rate
- elimination rate
- volume of distribution

HILL PD MODEL

$$E(C) = \frac{E_{max} \cdot C^{Y}}{+\,C^{Y}}\,(EC_{50}$$

- pharmacodynamic effect
- maximum effect
- concentration for 50% effect
- hill coefficient

12-HOUR IN-SILICO COMPARISON

Metoprolol 25 mg	
Metoprolol 50 mg	
Ivabradine 5 mg	
Amiodarone 200 mg	

BRADYCARDIA GUARDRAIL | model flags low heart rate risk across simulated regimens

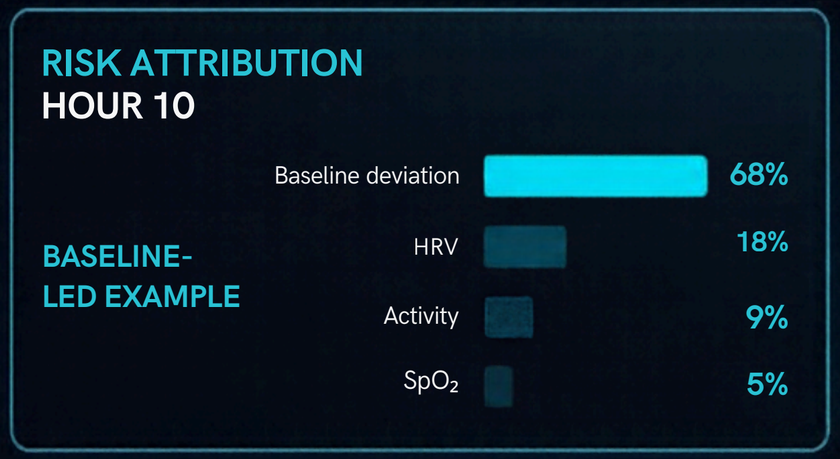
SIMULATION ONLY | not clinically validated. not treatment advice.

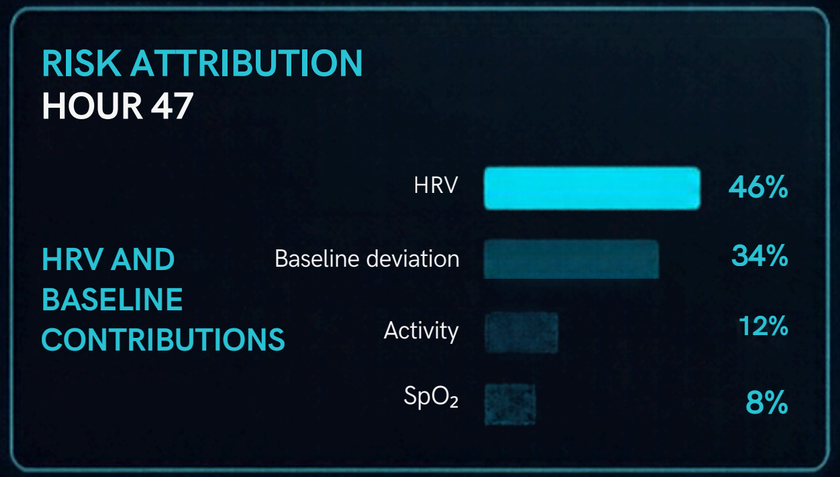
Explainability & Escalation

Risk drivers are transparent and shift over time.

RED status initiates audit logging.

Clinicians are alerted via webhook in **real time.**





Audit log

Time (UTC)	Event	
2024-05-21 14:47:12	RED STATUS TRIGGERSD	Risk score 0.87
2024-05-21 14:47:12	RISK_SNAPSHOT	Attribution captured
2024-05-21 14:47:13	AUDIT_LOGGED	Record ID 712c...
2024-05-21 14:47:14	WEBHOOK_SENT	Clinician alert

Webhook payload


```
"patient_id"; "PT-12845"; "event";
"REDSTATUS TRIGGered", "risk_score"; 8 .87,
"timestamp"; "2024-05-21T14:47:122";
"reason"; "riskthresholdExceeded"; "action";
"reviewRequired"
```

FHIR R4 & P4 Companion

Seamless hospital interoperability.
A complete, empathetic patient view.

Unified Patient, Observation and RiskAssessment bundle

LOINC-coded observations for standardized data exchange

Future-ready integration with ABDM ecosystem

ABDM INTEGRATION

Future target

Prototype Validation Metrics

94%	accuracy
0.995	ROC-AUC
15-hour	simulated lead time
26.4%	simulated risk reduction
<15 ms	latency

Synthetic sandbox results and investigative simulations.
Not clinically validated. Not prescribing guidance.

Roadmap to Clinical Translation

SYNTHETIC SANDBOX

Proof of concept in synthetic environments

LIVE WEARABLES AND ECG MODELLING

Real-world signals and advanced modelling

ETHICS-APPROVED PROSPECTIVE PILOT

Governed deployment in prospective settings